Let $\phi : G \to G''$ be defined by $\phi(a) = \rho_{a^{-1}}$. Clearly ϕ is one to one and maps G onto G''. From the equation $\rho_a\rho_b = \rho_{ba}$ derived above, we have $\phi(ab) = \rho_{(ab)^{-1}} = \rho_{b^{-1}a^{-1}} = \rho_{a^{-1}}\rho_{b^{-1}} = \phi(a)\phi(b)$, which is the homomorphism property for ϕ. Therefore ϕ is an isomorphism of G onto G''.

53. a. Let us show that that the $n \times n$ permutation matrices form a subgroup of the group $GL(n, \mathbb{R})$ of all invertible $n \times n$ matrices under matrix multiplication. If P_1 and P_2 are two of these permutation matrices, then the exercise stated that P_1P_2 is the matrix that produces the same reordering of the rows of P_2 as the reordering of the rows of I_n that produced P_1. Thus P_1P_2 can again be obtained from the identity matrix I_n by reordering its rows, so it is a permutation matrix. The matrix I_n is the identity permutation matrix. If P is obtained from I_n by a reordering the rows that puts row i in the position j, then P^{-1} is the matrix obtained from I_n by putting row j in position i. Thus the $n \times n$ permutation matrices do form a group under permutation multiplication.

Let us number the elements of G from 1 to n, and number the rows of I_n from 1 to n, say from top to the bottom in the matrix. Theorem 8.16, says we can associate with each $g \in G$ a permutation (reordering) of the elements of G, which we can now think of as a reordering of the numbers from 1 to n, which we can in turn think of as a reordering of the rows of the matrix I_n, which is in turn produced by multiplying I_n on the left by a permutation matrix P. The effect of left multiplication of a matrix by a permutation matrix, explained in the exercise, shows that this association of g with P is an isomorpism of G with a subgroup of the group of all permutation matrices.

b. Proceeding as in the second paragraph of Part(**a**), we number the elements e, a, b, and c of the Klein 4-group in Table 5.11 with the numbers 1, 2, 3, and 4 respectively. Looking at Table5.11, we see that left multiplication of each of e, a, b, c by a produces the sequence a, e, c, b. Applying the same reordering to the numbers 1, 2, 3, 4 produces the reordering 2, 1, 4, 3. Thus we associate with a the matrix obtained from the I_4 by interchanging rows 1 and 2 and interchanging rows 3 and 4. Proceeding in this fashion with the other three elements, we obtain these pairings requested in the exercise.

$$e \leftrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad a \leftrightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad b \leftrightarrow \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad c \leftrightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

9. Orbits, Cycles, and the Alternating Groups

1. $\{1, 2, 5\}, \{3\}, \{4, 6\}$ **2.** $\{1, 5, 7, 8\}, \{2, 3, 6\}, \{4\}$ **3.** $\{1, 2, 3, 4, 5\}, \{6\}, \{7, 8\}$ **4.** $\mathbb{Z}$

5. $\{2n \mid n \in \mathbb{Z}\}, \{2n + 1 \mid n \in \mathbb{Z}\}$ **6.** $\{3n \mid n \in \mathbb{Z}\}, \{3n + 1 \mid n \in \mathbb{Z}\}, \{3n + 2 \mid n \in \mathbb{Z}\}$

7. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 3 & 5 & 8 & 6 & 2 & 7 \end{pmatrix}$ **8.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 7 & 2 & 8 & 5 & 4 & 1 & 6 \end{pmatrix}$ **9.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 3 & 7 & 8 & 6 & 2 & 1 \end{pmatrix}$

10. (1,8),(3, 6, 4)(5, 7) and (1, 8)(3, 4)(3, 6)(5, 7)

11. (1, 3, 4)(2, 6)(5, 8, 7) and (1, 4)(1, 3)(2, 6)(5, 7)(5, 8)

12. (1, 3, 4, 7, 8, 6, 5, 2) and (1, 2)(1, 5)(1, 6)(1, 8)(1, 7)(1, 4)(1, 3)

13. (See the answer in the text.)

14. The greatest order is 6 and comes from a product of disjoint cycles of lengths 2 and 3.

15. The greatest order is 6 and comes from a cycle of length 6.

16. The greatest order is 12, coming from a product of disjoint cycles of lengths 4 and 3.

17. The greatest order is 30 and comes from a product of disjoint cycles of lengths 2, 3, and 5.

18. The greatest order is 105 and comes from a product of disjoint cycles of lengths 3, 5, and 7.

19. (See the text answer.) **20.** The definition is correct.

21. The definition is incorrect; (1,4,5) is a cycle in S_5, but it has three orbits, $\{1,4,5\}, \{2\}$, and $\{3\}$.

A permutation σ of a finite set is a **cycle** if and only if σ has at most one orbit of cardinality greater than 1.

22. The definition is incorrect; it must be specified as a subgroup of some S_n.

The **alternating group** A_n is the subgroup of S_n consisting of the even permutations in S_n.

23. F T F F F F T T T F

24. The even permutations in S_3 are $\rho_0 = (12)(12), \rho_1 = (1,2,3) = (1,3)(1,2)$, and $\rho_2 = (1,3,2) = (1,2)(1,3)$.

	ρ_0	ρ_1	ρ_2
ρ_0	ρ_0	ρ_1	ρ_2
ρ_1	ρ_1	ρ_2	ρ_0
ρ_2	ρ_2	ρ_0	ρ_1

25. Viewing a permutation σ in S_n as permuting the rows of the identity matrix I_n, we see that if σ could be expressed as both an even and odd number of transpositions (giving row interchanges), then the matrix resulting from applying σ to I_n would have both determinant 1 and determinant -1.

26. If σ is a permutation and $\tau = (i,j)$ is a transposition in S_n, then by considering whether i and j are in the same or different orbits of σ, we can show that the number of orbits of σ and of $\tau\sigma$ differ by 1. Starting with the identity permutation ι which has n orbits and multiplying by transpositions to produce σ, we see that the number of transpositions can't be both even and odd, for σ has either an even or odd number of orbits, but not both.

27. **a.** Note that (1, 2)(1, 2) is the identity permutation in S_n, and $2 \leq n-1$ if $n > 2$. Because $(1,2,3,4,\cdots,n) = (1,n)(1,n-1)\cdots(1,3)(1,2)$, we see that a cycle of length n can be written as a product of $n-1$ transpositions. Now a permutation in S_n can be written as a product of disjoint cycles, the sum of whose lengths is $\leq n$. If there are r disjoint cycles involved, we see the permutation can be written as a product of at most $n-r$ transpositions. Because $r \geq 1$, we can always write the permutation as a product of at most $n-1$ transpositions.

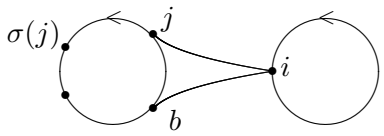
b. This follows from our proof of **a.**, because we must have $r \geq 2$.

c. Write the odd permutation σ as a product of s transpositions, where $s \leq n-1$ by Part(**a**). Then s is an odd number and $2n+3$ is an odd number, so $2n+3-s$ is an even number. Adjoin $2n+3-s$ transpositions (1,2) as factors at the right of the product of the s transpositions that comprise σ. The same permutation σ results because the product of an even number of factors (1,2) is the identity permutation. Thus σ can be written as a product of $2n+3$ permutations.

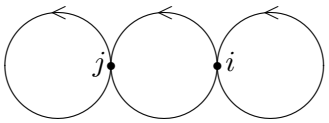
If σ is even, we proceed in exactly the same way, but this time s is even so $2n + 8 - s$ is also even. We tack the identity permutation, written as a product of the $2n + 8 - s$ factors (1, 2), onto the end of σ and obtain σ as a product of $2n + 8$ transpositions.

28. LaTeX is unable to draw these figures the way I would like. Make the modifications listed in your own sketches. The final solid lines in your sketch will indicate the orbit after performing the additional transposition (i, j).

a. Consider the right circle to be drawn with a dashed rather than solid curve, and also the short arc from b to j on the left circle to be dashed.



b. Consider the left and right circles both to be drawn with dashed curves, indicating the orbits before performing the additional transposition (i, j).



29. Suppose $\sigma \in H$ is an odd permutation. Let $\phi : H \to H$ be defined by $\phi(\mu) = \sigma\mu$ for $\mu \in H$. If $\phi(\mu_1) = \phi(\mu_2)$, then $\sigma\mu_1 = \sigma\mu_2$, so $\mu_1 = \mu_2$ by group cancellation. Also, for any $\mu \in H$, we have $\phi(\sigma^{-1}\mu) = \sigma\sigma^{-1}\mu = \mu$. This shows that ϕ is a one-to-one map of H onto itself. Because σ is an odd permutation, we see that ϕ maps an even permutation onto an odd one, and an odd permutation onto an even one. Because ϕ maps the set of even permutations in H one to one onto the set of odd permutations in H, it is immediate that H has the same number of even permutations as odd permutations. Thus we have shown that if H has one odd permutation, it has the same number of even permutations as odd permutations.

30. If the cycle has length 1, then no element is moved. If it has length $n > 1$, then n elements are moved, because elements not in the cycle are not moved.

31. *Closure:* Let $\sigma, \mu \in H$. If σ moves elements $s_1, s_2, \cdots, s_k$ of A and μ moves elements $r_1, r_2, \cdots, r_m$ of A, then $\sigma\mu$ can't move any elements not in the list $s_1, s_2, \cdots, s_k, r_1, r_2, \cdots, r_m$, so $\sigma\mu$ moves at most a finite number of elements of A, and hence is in H. Thus H is closed under the operation of S_A.

Identity: The identity permutation is in H because it moves no elements of A.

Inverses: Because the elements moved by $\sigma \in H$ are the same as the elements moved by σ^{-1}, we see that for each $\sigma \in H$, we have $\sigma^{-1} \in H$ also. Thus H is a subgroup of S_A.

32. No, K is not a subgroup. If $\sigma, \mu \in K$ and σ is a cycle of length 40 while μ is a cycle of length 30 and these two cycles are disjoint, then $\sigma\mu$ moves 70 elements of A, and is thus not in K. Thus K is not closed under permutation multiplication.

33. Let μ be any odd permutation in S_n. Because σ is an odd permutation, so is σ^{-1}, and consequently $\sigma^{-1}\mu$ is an even permutation, and thus is in A_n Because $\mu = \sigma(\sigma^{-1}\mu)$, we see that μ is indeed a product of σ and a permutation in A_n.

34. It is no loss of generality to assume that $\sigma = (1, 2, 3, \cdots, m)$ where m is odd. Because m is odd, we easily compute that

$$\sigma^2 = (1, 2, 3, \cdots, m)(1, 2, 3, \cdots, m)(1, 3, 5, \cdots, m, 2, 4, 6, \cdots, m-1),$$

which is again a cycle.

35. If σ is a cycle of length n, then σ^r is also a cycle if and only if n and r are relatively prime, that is, if and only if $\gcd(n, r) = 1$. To see why, let the cycle be $\sigma = (1, 2, 3, \cdots, n)$ Computing, we find that σ^r carries 1 into $1 + r$, or more precisely, into $1 + r$ modulo n in case $r \geq n$. Then $1 + r$ modulo n is carried in turn into $1 + 2r$ modulo n, etc. Thus the cycle in σ^r containing 1 is

$$(1, 1 + r, 1 + 2r, 1 + 3r, \cdots, 1 + mr)$$

where all entries are to be read modulo n, and m is the smallest positive integer such that $1 + mr \equiv 1 (\text{mod } n)$, or equivalently, the smallest positive integer such that $mr \equiv 0 (\text{mod } n)$. Thus the length of the cycle containing 1 in σ^r is the smallest positive integer m such that mr is divisible by n. In order for σ^r to be a cycle, this value of m must be n, which is the case if and only if $\gcd(n, r) = 1$.

36. We must show that λ_a is one to one and onto G. Suppose that $\lambda_a(g_1) = \lambda_a(g_2)$. Then $ag_1 = ag_2$. The group cancellation property then yields $g_1 = g_2$, so λ_a is one to one. Let $g \in G$. Then $\lambda_a(a^{-1}g) = a(a^{-1}g) = g$, so λ_a is onto G.

37. *Closure:* Let $\lambda_a, \lambda_b \in H$. For $g \in G$, we have $(\lambda_a\lambda_b)(g) = \lambda_a(\lambda_b(g)) = \lambda_a(bg) = (ab)g = \lambda_{ab}(g)$. Thus $\lambda_a\lambda_b = \lambda_{ab}$, so H is closed under permutation multiplication (function composition).

Identity: Cearly λ_e is the identity permutation of G.

Inverses: We have $\lambda_a\lambda_{a^{-1}} = \lambda_{aa^{-1}} = \lambda_e$, so $\lambda_a^{-1} = \lambda_{a^{-1}}$. Thus H is a subgroup of S_G.

38. We must show that for each $a, b \in G$, there exists some $\lambda_c \in H$ such that $\lambda_c(a) = b$. We need only choose c such that $ca = b$. That is, we take $c = ba^{-1}$.

39. We show that $(1, 2, 3, \cdots, n)^r(1, 2)(1, 2, 3, \cdots, n)^{n-r} = (1, 2)$ for $r = 0$, (2, 3) for $r = 1$, (3, 4) for $r = 2, \cdots,$ $(n, 1)$ for $r = n - 1$. To see this, note that any number not mapped into 1 or 2 by $(1, 2, 3, \cdots, n)^{n-r}$ is left fixed by the given product. For $r = i$, we see that $(1, 2, 3, \cdots, n)^{n-i}$ maps $i+1$ into 1, which is then mapped into 2 by (1, 2), which is mapped into $i+2$ by $(1, 2, 3, \cdots, n)^i$. Also $(1, 2, 3, \cdots, n)^{n-i}$ maps $i + 2$ mod n into 2, which is then mapped into 1 by (1, 2), which is mapped into $i + 1$ by $(1, 2, 3, \cdots, n)^i$.

Let (i, j) be any transposition, written with $i < j$. We easily compute that

$$(i, j) = (i, i+1)(i+1, i+2)\cdots(j-2, j-1)(j-1, j)(j-2, j-1)\cdots(i+1, i+2)(i, i+1).$$

By Corollary 9.12, every permutation in S_n can be written as a product of transpositions, which we now see can each be written as a product of the special transpositions $(1, 2), (2, 3), \cdots, (n, 1)$ and we have shown that these in turn can be expressed as products of (1, 2) and $(1, 2, 3, \cdots, n)$. This completes the proof.